



OHDSI Example Study Protocol

Background

Participation in collaborative OHDSI network studies requires more than access to observational health data standardized to the OMOP Common Data Model (CDM). Sites must also demonstrate the ability to execute standardized study packages, generate cohorts, run analytical workflows, and produce study outputs using the OHDSI software ecosystem.

Although many organizations have successfully transformed their data to the OMOP CDM, the ability to reliably execute OHDSI studies varies across sites due to differences in technical infrastructure, software environments, local governance processes, and operational experience. As a result, there is a need for a standardized approach that allows sites to assess their readiness to participate in future network studies.

This protocol describes a sample OHDSI study designed to evaluate study execution capabilities rather than answer a scientific research question. The study package includes representative cohorts and analyses commonly used within the OHDSI community and is intended to exercise core components of the OHDSI analytical workflow.

The analytical methods included in this protocol were selected not to address a specific scientific question, but to exercise a broad range of standardized HADES capabilities that are commonly used in OHDSI network studies, thereby providing a comprehensive assessment of site readiness and study execution capability.

Consistent with the principles of transparency and reproducibility embodied within the OHDSI ecosystem, study outputs are generated as standardized delimited text files (CSV format) that can be independently reviewed by participating sites. This enables sites to understand the information produced by each analysis and inspect the results that would typically be shared in a distributed network study.

The primary outcome of interest is successful execution of the study package. Analytical results generated by participating sites are intended solely for local validation purposes and are not intended to be interpreted as scientific findings, compared across sites, pooled, or submitted to a coordinating center.

Intended Use of Results

Results generated from this protocol are intended solely for local validation of study execution capabilities. Results should not be interpreted as scientific findings, compared across sites, pooled across data partners, or submitted to a coordinating center.

The primary outcome of interest is successful execution of the study workflow, including cohort generation, analytical execution, and output production.

Study Objectives

Primary Objective

To assess a site's ability to successfully execute a standardized OHDSI study package using data mapped to the OMOP Common Data Model.

Secondary Objectives

- Verify successful cohort generation.
 - Verify successful execution of standardized OHDSI analytical workflows.
 - Verify successful generation of study outputs.
 - Identify technical or operational issues that may impede participation in future OHDSI network studies.
 - Provide sites with experience executing a complete OHDSI study workflow.
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Study Design

This is a cross-sectional site readiness assessment conducted using observational health data mapped to the OMOP Common Data Model.

Participating sites will execute a standardized study package developed using the OHDSI Health Analytics Data-to-Evidence Suite (HADES) and specified using the Strategus study specification framework.

The analyses included in this protocol are intended to validate study execution capabilities and are not designed to address a specific clinical, epidemiological, or methodological research question.

Data Source Requirements

Participating sites should execute the study using one or more databases standardized to the OMOP Common Data Model.

Databases should satisfy the following requirements:

- Compatible OMOP CDM version.
- Required vocabularies available.
- Sufficient permissions to create cohorts and execute analyses.
- Required software dependencies installed.
- Adequate computational resources available.

No patient-level data, aggregate results, analytical estimates, or study outputs are required to be shared outside the participating institution.

Cohort Definitions

Primary Cohorts

Cohort ID	Cohort Name	Purpose
20126	ACE inhibitor	Exposure cohort
20127	Diuretic	Comparator exposure cohort
20128	Hypertensive disorder	Indication cohort
20129	Acute myocardial infarction	Outcome cohort
20130	Angioedema	Outcome cohort

Derived Cohorts

Cohort ID	Cohort Name	Purpose
20126001	ACE inhibitor subset	Exposure cohort restricted to hypertensive patients meeting study eligibility criteria
20127001	Diuretic subset	Comparator cohort restricted to hypertensive patients meeting study eligibility criteria
20128101	Hypertensive disorder subset	Restricted indication cohort

The derived cohorts apply first-ever occurrence restrictions, observation-time requirements, demographic restrictions, and study-period restrictions between January 1, 2020 and December 31, 2024.

Analysis Specifications

Cohort Generation

All primary and derived cohorts are generated as part of the study package.

Cohorts Generated

Cohort ID	Cohort Name
20126	ACE inhibitor
20127	Diuretic
20128	Hypertensive disorder
20129	Acute myocardial infarction
20130	Angioedema
20126001	ACE inhibitor subset
20127001	Diuretic subset
20128101	Hypertensive disorder subset

Analysis Purpose

The cohorts above are generated to identify patients and are used as inputs for the subsequent analyses.

CohortDiagnostics

Cohorts Evaluated

Cohort ID	Cohort Name	Purpose
20126	ACE inhibitor	Exposure cohort
20127	Diuretic	Comparator cohort

20128	Hypertensive disorder	Indication cohort
20129	Acute myocardial infarction	Outcome cohort
20130	Angioedema	Outcome cohort
20126001	ACE inhibitor subset	Derived exposure cohort
20127001	Diuretic subset	Derived comparator cohort
20128101	Hypertensive disorder subset	Derived indication cohort

Analysis Purpose

CohortDiagnostics is executed for all primary and derived cohorts to evaluate cohort construction, concept usage, incidence, cohort relationships, and temporal characteristics.

Characterization

Baseline Characterization

Target Cohort ID Target Cohort Name

20126	ACE inhibitor
20127	Diuretic
20128	Hypertensive disorder
20126001	ACE inhibitor subset
20127001	Diuretic subset
20128101	Hypertensive disorder subset

Time-to-Event Characterization

Target Cohorts

Target Cohort ID Target Cohort Name

20126	ACE inhibitor
20127	Diuretic
20128	Hypertensive disorder

20126001	ACE inhibitor subset
20127001	Diuretic subset
20128101	Hypertensive disorder subset

Outcomes

Outcome Cohort ID	Outcome Cohort Name
20129	Acute myocardial infarction
20130	Angioedema

Dechallenge/Rechallenge Characterization

Target Cohorts

Target Cohort ID	Target Cohort Name
20126	ACE inhibitor
20127	Diuretic
20128	Hypertensive disorder
20126001	ACE inhibitor subset
20127001	Diuretic subset
20128101	Hypertensive disorder subset

Outcomes

Outcome Cohort ID	Outcome Cohort Name
20129	Acute myocardial infarction
20130	Angioedema

Analysis Purpose

These analyses characterize baseline clinical characteristics, evaluate outcome occurrence following cohort entry, and assess temporal relationships between exposures and outcomes.

CohortIncidence

Target Populations

Target Cohort ID Target Cohort Name

20126001	ACE inhibitor subset
20127001	Diuretic subset
20128101	Hypertensive disorder subset

Outcomes

Outcome Cohort ID Outcome Cohort Name Clean Window

20129	Acute myocardial infarction	365 days
20130	Angioedema	365 days

Time-at-Risk Definition

Parameter Definition

Risk window start	1 day after target cohort start
Risk window end	Target cohort end date

Study Window and Stratification

Parameter Setting

Study period	2020-01-01 through 2024-12-31
Stratification	Age, gender, calendar year

Analysis Purpose

This component validates execution of standardized OHDSI incidence estimation workflows and generation of incidence-related outputs.

CohortMethod

Treatment Comparison

Role	Cohort ID	Cohort Name
Target	20126001	ACE inhibitor subset
Comparator	20127001	Diuretic subset
Indication Population	20128	Hypertensive disorder

The target and comparator cohorts are nested within the hypertensive disorder cohort, requiring treatment exposure to occur during a period of membership in the indication cohort.

Outcomes of Interest

Outcome Cohort ID	Outcome Cohort Name
20129	Acute myocardial infarction
20130	Angioedema

Time-at-Risk

Parameter	Definition
Risk window start	1 day after treatment initiation
Risk window end	End of treatment exposure
Analysis type	On-treatment

Negative Controls

Additional negative control outcomes are included to support empirical calibration. Negative controls are outcomes believed not to be causally related to either treatment and are assigned a true effect size of 1.0.

Effect estimates will be calibrated using established OHDSI empirical calibration methods.

Analysis Purpose

This analysis validates execution of the OHDSI CohortMethod framework, including indication restriction, propensity score adjustment, comparative effect estimation, negative control evaluation, and empirical calibration.

Self-Controlled Case Series (SCCS)

Exposure and Outcome Definitions

Outcomes of Interest

Exposure Cohort	Outcome Cohort	Indication Population
ACE inhibitor	Acute myocardial infarction	Hypertensive disorder
ACE inhibitor	Angioedema	Hypertensive disorder
Diuretic	Acute myocardial infarction	Hypertensive disorder
Diuretic	Angioedema	Hypertensive disorder

Negative Controls

Additional negative control outcomes are included to support evaluation of residual systematic error. Negative control outcomes are assigned a true effect size of 1.0 and are detailed in Appendix A.

Risk Windows

Risk Period	Definition
Pre-exposure period	30 days before exposure start through 1 day before exposure start
Main exposure period	1 day after exposure start through exposure end

Study Population Construction

Parameter	Setting
Study period	2020-01-01 through 2024-12-31
Naïve period	365 days
First outcome only	Yes

The SCCS analyses are nested within the hypertensive disorder cohort and evaluate exposure periods occurring during membership in the indication cohort.

Analysis Purpose

This analysis validates execution of the OHDSI SCCS framework, including population construction, exposure era generation, risk-window specification, within-person effect estimation, and negative control evaluation.

Patient-Level Prediction

Prediction Problems

Target Cohort ID	Target Cohort Name	Outcome Cohort ID	Outcome Cohort Name
20126001	ACE inhibitor subset	20129	Acute myocardial infarction
20126001	ACE inhibitor subset	20130	Angioedema

Population Construction

Parameter	Setting
Study period	2020-01-01 through 2024-12-31
Prior observation requirement	365 days
Outcome exclusion	Prior outcome excluded
Prediction window start	1 day after cohort entry
Prediction window end	365 days after cohort entry
Minimum time at risk	364 days

Predictors are derived from demographic characteristics, clinical conditions, drug exposures, and healthcare utilization observed prior to cohort entry.

Analysis Purpose

This analysis validates execution of a complete OHDSI patient-level prediction workflow, including population construction, covariate generation, model training, and performance assessment.

Analytical Methods

The analyses included in this study package are implemented using standardized methods from the OHDSI Health Analytics Data-to-Evidence Suite (HADES).

Detailed descriptions of the analytical methods, methodological evaluations, and associated peer-reviewed publications are available through the HADES Methods Publications repository:

https://ohdsi.github.io/Hades/publications.html#Methods_research

Success Criteria

Infrastructure Readiness

- Study package installs successfully.
- Required dependencies load successfully.
- Connection to OMOP CDM is established.
- Required vocabularies are available.

Cohort Generation

- All primary and derived cohorts are generated.
- Cohort generation statistics are produced.
- No unrecoverable errors occur.

Analytical Execution

Successful completion of:

- Cohort Generation
- CohortDiagnostics
- Characterization
- CohortIncidence
- CohortMethod
- SCCS

- Patient-Level Prediction

Output Generation

Expected outputs are generated for each module, including:

- Cohort counts
- CohortDiagnostics results
- Characterization results
- Incidence estimates
- CohortMethod results
- SCCS results
- Patient-Level Prediction results

Output Transparency and Review

All outputs are written to standardized CSV files and can be inspected using text editors, spreadsheet software, statistical software, or database tools.

Sites are encouraged to review these files to understand the information that would typically be returned to a study coordinator in a distributed OHDSI network study.

Readiness Determination

A site may be considered ready to participate in OHDSI network studies when all analytical modules complete successfully and expected outputs are generated without critical execution errors.

Interpretation

Successful completion of this protocol demonstrates a site's ability to execute a representative OHDSI study package using data mapped to the OMOP Common Data Model.

The protocol is intended to assess technical readiness rather than generate evidence. Analytical results should not be interpreted as scientific findings and should not be compared across sites, pooled across organizations, or submitted to a study coordinator.

The primary measure of success is successful execution of the study workflow and generation of expected outputs.

Appendix

Appendix A

Negative control outcomes are included to support empirical calibration of effect estimates. These outcomes are believed not to be causally related to either treatment and are assigned an expected true effect size of 1.0. Negative control outcomes are used within the CohortMethod and Self-Controlled Case Series analyses to evaluate and adjust for residual systematic error. The full list of negative control outcomes used is found here:

<https://github.com/ohdsi-studies/OhdsiExampleStudy/blob/main/inst/negativeControlOutcomes.csv>